George Lakoff, Mark Johnson, Conceptual Metaphors, and Reality Drift

What Happens When Informational Systems Begin Structuring Perception at Scale

Intellectual Lineage Note #3

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The Basic Pattern

Human beings do not encounter reality as raw information. Experience is organized through categories, metaphors, frames, and recurring patterns that make complex conditions easier to understand. These structures allow people to reason about abstract ideas by relating them to more familiar forms.

George Lakoff and Mark Johnson identified the depth of this process through conceptual metaphor theory. Their central argument was that metaphor is not merely a feature of language. It is part of the structure of thought. People do not only describe time as money, arguments as war, or knowledge as vision. They often reason and act through those mappings.

This creates a permanent cognitive vulnerability. A framework can guide attention so effectively that its own role disappears from awareness. The person continues seeing through the representation without noticing that the representation is shaping what can be seen.

Modern informational systems extend this problem beyond language. Platforms, dashboards, rankings, profiles, benchmarks, and interfaces increasingly organize experience through representational structures of their own. They do not simply communicate information. They establish the forms through which people interpret value, capability, relevance, identity, and success.

Within the Reality Drift framework, conceptual metaphor therefore points toward a broader structural problem. What happens when representational systems become embedded in the environments through which individuals and institutions perceive reality?

Metaphor Is Not Decoration

Metaphors work by mapping structure from one domain onto another. An abstract condition becomes understandable through a more concrete pattern. Time can be spent. Ideas can be grasped. Arguments can be attacked. Relationships can move forward or fall apart.

These expressions are not random. They organize expectations, direct attention, and make some actions feel more natural than others. When an argument is understood as war, disagreement becomes a contest with positions, attacks, defenses, and winners. When time is understood as money, it becomes a resource that can be saved, wasted, invested, or lost.

The metaphor does not need to be consciously believed in order to influence reasoning. Its power often comes from becoming ordinary. Once embedded in language and practice, it can shape perception without appearing as an interpretation.

Lakoff and Johnson's work matters because it shows that representations do not enter thought after perception has already occurred. They help organize perception from the beginning. The framework determines which similarities matter, which relationships become visible, and which possibilities remain difficult to imagine.

Understanding the Problem

Conceptual Metaphor: A cognitive mapping through which one domain of experience is understood in terms of another.

Source Domain: The familiar or concrete structure used to organize understanding, such as movement, conflict, vision, possession, or exchange.

Target Domain: The more abstract condition being understood, such as time, knowledge, relationships, argument, or identity.

Cognitive Framing: The organization of perception and reasoning through recurring assumptions, categories, and representational structures.

Representational Environment: An informational setting in which interfaces, metrics, rankings, categories, and symbolic systems continuously shape how reality is interpreted.

Frame Optimization: The repeated adjustment of an environment around representations that produce measurable attention, compliance, engagement, or performance.

Perceptual Drift: The gradual narrowing of what individuals or institutions can recognize as representations increasingly determine which aspects of reality become visible.

These processes are not inherently distorting. Metaphors and frames make thought possible by reducing complexity into usable forms. The danger appears when a particular representation becomes so embedded that alternatives disappear from awareness and the framework begins to function as reality itself.

What Lakoff and Johnson Identified

Lakoff and Johnson argued that abstract thought is grounded in bodily experience. People understand complex ideas through recurring patterns involving movement, orientation, balance, containment, force, distance, and physical interaction.

This means cognition is not built from detached symbols alone. Meaning emerges through embodied relationships with the world. Up and down, inside and outside, near and far, forward and backward become foundations for reasoning about status, emotion, time, morality, and social organization.

Conceptual metaphors also highlight some aspects of experience while concealing others. Understanding argument as war makes conflict, strategy, and victory more visible. It makes cooperation, exploration, and mutual revision less central. No metaphor captures the whole target domain.

The problem is not that the metaphor is false. The problem is that its partial structure can become mistaken for the complete structure of the thing being understood.

Lakoff and Johnson identified how cognitive framing shapes experience before deliberate evaluation begins. Modern systems turn this process into an engineered environment.

From Conceptual Metaphors to Representational Systems

Conceptual metaphor theory explains how one structure organizes understanding of another. Modern systems extend this mechanism through representations that are built into platforms, institutions, and interfaces.

A dashboard presents an organization as a collection of measurable indicators. A follower count presents influence as visible audience size. A credential presents capability as certified attainment. A benchmark presents intelligence as performance on a standardized evaluation. A recommendation feed presents relevance as predicted engagement.

These are not always metaphors in the narrow linguistic sense. They function similarly by carrying structure from one domain into another. The measurable becomes a model for the meaningful. The ranked becomes a model for the valuable. The visible becomes a model for the real.

Once embedded in everyday systems, these representations stop appearing as interpretations. They become the environment through which interpretation occurs. The question is therefore no longer only how metaphors shape thought. It is how entire systems can stabilize particular frames, reward behavior that confirms them, and make alternative interpretations increasingly difficult to sustain.

How Representational Framing Becomes Structural

Stage 1 — Experience

An individual or institution encounters a complex condition such as learning, influence, competence, risk, public opinion, or organizational health.

Stage 2 — Framing

The condition is organized through a metaphor, category, model, metric, interface, or symbolic structure. Some features become central while others recede.

Stage 3 — Institutionalization

The frame becomes embedded in procedures, technologies, incentives, and shared language. It no longer operates only as an interpretation. It begins guiding decisions.

Stage 4 — Adaptation

People adjust behavior to the structure made visible by the frame. They learn what counts, what receives attention, and which forms of action remain legible.

Stage 5 — Environmental Reinforcement

The system amplifies behavior that fits the frame. New data, content, and decisions are generated within the same structure, making the frame appear increasingly accurate.

Stage 6 — Perceptual Drift

The representation becomes difficult to distinguish from the reality it organizes. Alternative dimensions of the condition become less visible, and the system loses awareness of what its frame has excluded.

The Framing Stack

Modern perception often passes through several representational layers:

Experience → Metaphor → Category → Interface → Metric → Ranking → Decision → Behavior → New Experience

A complex condition is first interpreted through a recurring metaphor or frame. That frame becomes formalized through categories and interfaces. The categories generate metrics, and the metrics produce rankings or evaluations. These representations guide decisions, decisions alter behavior, and the resulting behavior creates new experiences that reenter the same structure.

This forms a recursive loop. At first, the frame helps interpret reality. Over time, people and institutions begin reorganizing reality around the frame. What fits the representation becomes easier to see, reward, and reproduce. What does not fit becomes less visible or is treated as noise.

The system then encounters an environment already shaped by earlier acts of framing. A platform optimized for engagement generates content designed for engagement, then treats the resulting behavior as evidence that engagement accurately represents what users value. An organization governed by dashboards generates work designed for dashboards, then interprets the improved metrics as proof that performance has improved.

This is how a cognitive structure becomes an operating structure. The representation no longer only shapes thought about reality. It begins shaping the reality that later thought will interpret.

The Drift Principle

Within the Reality Drift framework, this process can be described through the Drift Principle. When representational systems increasingly organize perception, individuals and institutions gradually lose awareness of the distinction between the frame and the reality it makes visible.

The principle does not suggest that framing can be eliminated. All thought depends on selection, abstraction, and structure. The vulnerability appears when a frame becomes dominant enough to hide its own limits.

A metric can be easier to compare than the condition it represents. A dashboard can be easier to manage than the organization behind it. A ranking can be easier to trust than the judgment required to interpret it. As these representations become more stable and influential, they can displace direct attention to the realities they simplify.

The frame survives because the system continues producing information in its terms. The representation remains coherent because new behavior adapts to its categories. What becomes weaker is the ability to recognize what the framework no longer allows the system to see.

Examples Across Modern Representational Systems

Social Media Metrics

Follower counts, likes, views, and shares represent different forms of visibility and response. They provide useful signals about distribution and audience activity.

Once these metrics become the dominant frame for interpreting influence, visibility begins to stand in for importance. Content is then produced for measurable reaction, and the informational environment increasingly confirms the idea that attention is the most relevant form of value.

Personal Branding

Personal branding translates identity into a coherent public representation. It makes expertise, values, and professional direction easier to communicate.

The same structure can cause the person to interpret themselves through the brand. Experiences become valuable when they reinforce the narrative, while contradictions or changes become harder to integrate. Identity is increasingly understood through the metaphor of a marketable product.

Organizational Dashboards

Dashboards frame organizations as collections of measurable variables. They simplify complexity and help coordinate decisions across large systems.

As management becomes organized around the dashboard, measurable activity gains priority over less legible forms of judgment, care, maintenance, and long-term development. The organization begins behaving as though what appears on the dashboard is what the organization is.

Credentials and Capability

Credentials provide a visible representation of training, attainment, and institutional recognition. They allow capability to be inferred without direct evaluation in every interaction.

When credentials become the dominant frame, certified attainment can begin substituting for practical judgment or actual performance. The symbol continues carrying authority even as the relationship between the credential and capability becomes harder to assess.

AI Benchmarks

Benchmarks frame model capability through standardized tasks and numerical scores. They make comparison possible and provide shared signals of progress.

When the benchmark becomes the primary image of intelligence, development can narrow around what the evaluation recognizes. The score shapes how capability is perceived, and the field may begin treating improvement inside the frame as improvement in intelligence itself.

Recommendation Systems

Recommendation systems frame relevance as predicted response. They infer what matters from previous interaction and reorganize the informational environment around those predictions.

Users then encounter a world structured by what the system expects them to engage with. Their behavior adapts to that environment, producing new data that appears to confirm the original frame.

Time Is Money and the Optimization of Experience

Lakoff and Johnson's analysis of "time is money" shows how a conceptual metaphor can become embedded in institutional life. Time is treated as a resource that can be budgeted, spent, saved, wasted, invested, or lost.

The metaphor is useful because time is limited and activities require coordination. Yet it also highlights economic allocation while concealing other ways of understanding duration. Time can be lived, endured, shared, remembered, inhabited, or allowed to unfold without producing an output.

Modern optimization systems intensify the economic frame. Calendars, productivity software, time tracking, performance systems, and platform analytics convert more parts of experience into measurable units. The metaphor becomes embedded in the tools through which daily life is organized.

At that point, people do not merely speak as though time were money. They encounter environments that require them to manage time as an economic asset. Unmeasured time begins to feel wasted, and experience becomes increasingly evaluated through its visible return. The metaphor has become infrastructure.

Knowledge Is Seeing and the Interface of Understanding

Another recurring conceptual metaphor treats knowledge as vision. People seek insight, clarity, perspective, focus, illumination, and points of view. Ignorance appears as blindness, confusion as obscurity, and understanding as seeing clearly.

Digital systems often intensify this metaphor by making information visually available through dashboards, graphs, maps, and interfaces. What can be displayed appears understandable. What cannot be rendered remains difficult to evaluate.

This creates a subtle substitution. Visibility begins to stand in for knowledge. A system may provide more data, cleaner graphics, and more complete dashboards while weakening the contextual understanding required to interpret what is shown. The interface creates the feeling of seeing the system. Yet seeing a representation is not the same as understanding the reality that produced it.

Reality Drift emerges when informational clarity increases at the level of presentation while contact with the underlying condition becomes weaker.

Argument Is War and Algorithmic Conflict

The metaphor "argument is war" structures disagreement through positions, attacks, defenses, strategies, and victories. It makes adversarial reasoning intuitive while placing less emphasis on joint inquiry or mutual revision.

Social platforms amplify this frame because conflict often produces visible engagement. Strong opposition generates replies, shares, reactions, and continued attention. The platform does not need to explicitly instruct users to treat argument as war. Its incentives strengthen the forms of discourse already organized through that metaphor.

Participants then adapt. Opinions become positions to defend, uncertainty appears as weakness, and changing one's mind resembles surrender. The result is an environment that repeatedly confirms the adversarial frame by rewarding behavior shaped by it. A conceptual metaphor that once organized language becomes fused with an optimization system. The frame generates the behavior, the behavior generates engagement, and the engagement reinforces the frame.

Embodied Cognition and Digital Abstraction

Lakoff and Johnson emphasized that meaning is grounded in bodily experience. Abstract thought draws on patterns learned through movement, orientation, containment, force, balance, and physical interaction.

Digital systems often place additional layers of abstraction between cognition and the world. People encounter events through feeds, metrics, summaries, notifications, profiles, and synthetic media. The representation may remain visually immediate while the embodied conditions behind it become distant.

This does not mean digital understanding is unreal. It means the relationship between symbolic representation and embodied experience requires active maintenance. A person can know more facts about an event while having less contact with the conditions that give those facts significance.

As information travels through increasingly abstract systems, cognitive fluency can increase while experiential grounding weakens. The representation feels easy to process because it fits familiar frames. The underlying reality may remain complex, resistant, and difficult to inhabit. The system therefore rewards what can be made cognitively legible, not necessarily what preserves contact with the embodied world.

Semantic Fidelity and Conceptual Framing

Semantic Fidelity asks whether meaning survives as information is represented, compressed, transmitted, interpreted, and reused. Conceptual metaphor theory reveals that meaning depends not only on the information preserved, but also on the structure through which that information is understood.

A statement may remain factually accurate while being reframed through a metaphor that changes its significance. An organization can be described as a machine, a family, a market, or an organism. Each frame highlights different relationships, responsibilities, and possible actions.

Semantic fidelity therefore requires more than preserving words or facts. It requires preserving the assumptions, relationships, and limits that allow a representation to be interpreted appropriately.

Frame drift occurs when information passes into a new representational structure without preserving awareness of the original context. The content survives, but the cognitive organization changes. The same facts begin supporting a different understanding because the frame has altered what those facts appear to mean.

Constraint Collapse and Invisible Frames

Frames remain useful when they are constrained by realities that can resist them. A metaphor should organize understanding without becoming the only structure through which the target domain can be interpreted.

- A dashboard must remain constrained by conditions it cannot display.
- A benchmark must remain constrained by capabilities outside the evaluation.
- A profile must remain constrained by the person it represents.
- A metric must remain constrained by outcomes beyond the measurement.
- A recommendation system must remain constrained by forms of relevance not visible through engagement.

Constraint Collapse occurs when these external sources of correction weaken. The frame becomes increasingly self-validating because the system generates behavior and data in the same terms the frame recognizes.

The dashboard is confirmed by the activity organized around the dashboard. Engagement is confirmed by content produced for engagement. Rankings are confirmed by choices made in response to the rankings. The system appears to discover reality while increasingly producing the reality it knows how to discover.

Reality Drift as a General Principle

Lakoff and Johnson showed that thought is structured by metaphors and embodied frames before conscious reasoning begins. Their work undermined the idea that people first perceive a neutral reality and only later describe it through language.

Modern representational systems extend this insight into institutions and technologies. Categories, interfaces, metrics, rankings, and algorithms increasingly shape the environments within which perception occurs. The frame is no longer located only in thought. It is built into the system.

Reality Drift emerges when these frameworks become so influential that their partial view becomes difficult to recognize as partial. The system grows increasingly effective at producing information, behavior, and decisions that fit the frame. Internal coherence rises because each layer operates through the same distinctions.

What weakens is contact with aspects of reality that the representation cannot express. The system does not need to suppress them directly. It can simply make them less visible, less measurable, and less actionable.

Lakoff and Johnson showed that metaphors shape how reality is understood. Reality Drift examines what happens when those structures become optimized environments that increasingly shape what reality is allowed to become.

Related Concepts

Conceptual Metaphor: A cognitive mapping that structures one domain of experience through another.

Embodied Cognition: The view that meaning and reasoning are grounded in bodily experience, perception, movement, and interaction.

Cognitive Framing: The organization of attention and interpretation through recurring assumptions, categories, and representational structures.

Representational Environment: A setting in which symbolic systems, metrics, interfaces, and algorithms continuously shape perception and action.

Frame Drift: The gradual change in meaning that occurs when information moves into a different cognitive or representational structure.

Semantic Fidelity: The degree to which meaning survives representation, compression, transmission, interpretation, and reuse.

Recursive Compression: The repeated reduction of already simplified representations, causing distinctions to disappear while coherence remains.

Constraint Collapse: The loss of external realities capable of limiting or correcting a dominant frame.

Reality Drift: The condition in which systems remain operational while their representations gradually lose alignment with real-world conditions.

Frequently Asked Questions

Is Reality Drift another form of conceptual metaphor theory?

No. Conceptual metaphor theory explains how metaphorical mappings structure cognition and meaning. Reality Drift examines the broader condition in which representations, models, metrics, and frames gradually lose alignment with the realities they organize while the surrounding system continues functioning.

Are all metaphors distortions?

All metaphors highlight some features of a target domain while concealing others. This does not make them useless or inherently misleading. Problems emerge when one metaphor becomes dominant and its partial structure is mistaken for the whole condition.

How are metrics similar to conceptual metaphors?

Metrics map complex realities onto simplified numerical forms. Like metaphors, they preserve selected relationships while excluding others. Once embedded in decisions and incentives, they can also shape how the underlying reality is perceived and acted upon.

What is a representational environment?

A representational environment is a setting in which interfaces, rankings, profiles, categories, metrics, and algorithms do more than display information. Together, they establish the structures through which people interpret and respond to reality.

Can a frame become more accurate over time?

Yes. Frames can be revised through evidence, comparison, and direct contact with changing conditions. Drift begins when the system becomes better at reinforcing the frame than testing its limits.

Why do dominant frames become difficult to notice?

A successful frame becomes part of ordinary perception. People stop experiencing it as one interpretation among others and begin using it as the background structure through which alternatives are evaluated.

How are Lakoff and Johnson relevant to AI?

AI systems operate through learned representations, categories, embeddings, prompts, benchmarks, and interface conventions. These structures shape how information is selected, generated, and interpreted, often without remaining visible to users.

What prevents perceptual drift?

Frames need correction from observations, experiences, and interpretations they did not generate. Multiple models, qualitative judgment, embodied experience, and explicit attention to excluded conditions can help preserve awareness of a frame's limits.

Canonical References

Reality Drift: Concept Paper and Definition
A general account of how systems remain operational while gradually losing alignment with real-world conditions.

Semantic Fidelity: Concept Paper and Definition
A framework for evaluating whether meaning survives representation, compression, transmission, interpretation, and reuse.

Recursive Compression: Concept Paper and Definition
An explanation of how repeated abstraction can preserve coherence while removing distinctions and contextual structure.

Constraint Collapse: Concept Paper and Definition
An account of what happens when the external boundaries and real-world requirements constraining a system gradually weaken.

The Optimization Trap: Concept Paper and Definition
A model of how improvement against a selected objective can reduce alignment with the broader purpose the objective was meant to serve.

Related Search Terms

- George Lakoff
- Mark Johnson
- conceptual metaphor
- conceptual metaphor theory
- embodied cognition
- cognitive framing
- metaphor and thought
- source domain
- target domain
- representational systems
- representational environments
- frame drift
- semantic fidelity
- recursive compression
- constraint collapse
- algorithmic framing
- social media metrics
- AI benchmarks
- dashboard culture
- reality drift